



**The Impact of Sleep Quality, Sleep Deprivation, and Stress on
Academic Performance Among University Students**

**Second Year
2025 / 2026**

Prepared by:

Nadia Ahmed Fathy. 5240179

Hana mohamoud Abdelhafez Bakery 5240924

Mohreal Boutros Gabrial Fahim 5240226

Mohamed Ahmed Mostafa Elghonemy. 5241136

Nour Mohamed Hassan. 5241008

Submitted to:

Dr: sara osama

Abstract:

Poor sleep and high stress are common problems for university students, and they can easily ruin both well-being and academic success. To build better support systems for students, we need to understand how these factors truly connect.

This study aimed to look at how sleep quality, sleep duration, and study habits relate to students' grades and their stress levels.

We did a cross-sectional study using data from an online survey filled out by university students. We cleaned and analyzed the data using Python (with Pandas, SciPy, and Statsmodels). To find patterns, we used Pearson Correlation for sleep and grades, Independent Samples t-tests for gender and sleep disorders, One-Way ANOVA for study timing, and Spearman Rank Correlation and Kruskal-Wallis H-test to examine stress-related factors. Finally, we tried a Random Forest Classifier to see if we could predict academic outcomes.

The analysis showed no real connection between sleep quality and grades ($r = -0.082$, $p = 0.413$). Also, Neither sleep duration nor study timing showed a significant relationship with stress levels ($r = -0.031$, $p = 0.761$ and $H = 1.403$, $p = 0.705$ respectively). We found a clear difference in stress between genders ($t = -2.566$, $p = 0.012$), with female students reporting higher stress levels. On the other hand, sleep disorders ($p = 0.984$) and the time of day students studied ($p = 0.757$) didn't make a meaningful difference in grades. Lastly, the machine learning model hit a low predictive accuracy of 23.81%.

The findings suggest that while sleep quality deeply impacts stress, it doesn't directly change a student's grades within this group. Gender also plays a big part in how students experience stress. Universities should focus more on mental health support—especially for female students—and teach students the value of good sleep to help them manage stress.

Keywords:

Sleep, students, stress, performance, insomnia, study

Introduction:

The relationship between sleep quality, stress, and academic performance among university students has become an important topic in recent years. University life often includes academic pressure, assignments, exams, and daily responsibilities, which may affect both students' mental and physical health. In many cases, these pressures can negatively influence sleep quality and increase overall stress levels.

This research also highlights the strong connection between mental and physical well-being and academic performance. At the same time, academic stress itself may contribute to sleep problems and higher levels of stress among students. Although there is no single pattern that applies to everyone, since circumstances and personal factors differ from one student to another, long-term physical and psychological exhaustion is likely to have a significant effect on academic performance.

Sleep is considered one of the most important factors in students' daily lives. Any disturbance in sleep patterns may affect a student's overall lifestyle, especially concentration, focus, and ability to perform academically. In addition, stress can have a noticeable impact on academic performance, as it may lead to distraction, reduced focus, and mistakes even when students are well prepared. Therefore, analyzing the relationship between sleep quality, sleep deprivation, stress, and academic performance can provide a better understanding of how these factors influence university students and their academic outcome.

Objectives :

Primary objectives:

To investigate the impact of sleep quality, sleep deprivation, and stress on academic performance among university students.

Secondary Objectives:

1. 2. 3. 4. 5. To assess sleep patterns and their relationship to academic performance among students from different academic years.

To measure stress levels and identify their main causes (exams, assignments, family/financial problems, social relationships, etc.) and evaluate their impact on academic performance.

To examine the relationship between sleep deprivation and concentration during lectures and identify associated symptoms (fatigue, anxiety, headaches, low motivation).

To analyze the correlation between stress, sleep quality, and study productivity across different demographic groups (gender, academic year). To determine the most influential factors affecting academic performance (sleep quality, stress, time management, social media, physical health) among students.

Research Question

1. Is there a relationship between sleep quality and academic performance?
 - IV: Sleep quality
 - DV: Academic performance
2. What factors (such as sleep quality, and study habits) affect students' stress levels?
 - IVs: Sleep quality, study habits
 - DV: Stress level
3. Do sleep disorders affect students' academic performance?
 - IV: Sleep disorders
 - DV: Academic performance
4. What are the most common symptoms associated with sleeping fewer hours?
 - IV: Hours of sleep
 - DV: Sleep-related symptoms
5. Is there a relationship between study timing and academic performance?
 - IV: Study timing
 - DV: Academic performance
6. Is there a relationship between gender and stress level among students?
 - IV: Gender
 - DV: Stress level

Data Description:

The dataset used in this study was collected through an online questionnaire distributed to university students to investigate the relationship between sleep habits, stress levels, and academic performance. The survey included

demographic questions in addition to questions related to sleep quality, sleep duration, stress frequency, study habits, academic performance, sleep disorders, and symptoms associated with sleep deprivation.

The dataset contains both categorical and ordinal variables. Key variables included:

- Gender
- Average sleep duration per night
- Sleep quality impact on academic performance
- Frequency of stress during the semester
- Preferred study period
- Presence of insomnia or sleep disorders
- Academic performance evaluation
- Symptoms caused by insufficient sleep
- Students' opinions regarding stress effects and sleep improvement methods

The data was initially stored in CSV format and imported into Python for cleaning, preprocessing, and exploratory data analysis.

Libraries Used:

Several Python libraries were used throughout the project for data manipulation, preprocessing, statistical analysis, and visualization:

Library	Purpose
Pandas	Data loading, cleaning, manipulation, and analysis
NumPy	Numerical operations and array processing
Seaborn	Statistical data visualization
Matplotlib	Creating charts and visualizations
SciPy	Statistical analysis functions
Statsmodels	Descriptive statistical analysis

Statistical Tests used:

Research Question	Statistical Test
Sleep quality & academic performance	Spearman Correlation
Sleep duration & stress level	Spearman Correlation
Study Timing & stress level	Kruskal-Wallis Test
Sleep disorders & academic performance	Independent t-test
Sleep hours & symptoms	Chi-square Test
Study timing & academic performance	One-Way ANOVA
Gender & stress level	Independent t-test

Data Cleaning and Preprocessing:

Several preprocessing and data cleaning techniques were applied to improve the quality, consistency, and reliability of the dataset before conducting the analysis.

1. Data Importing

The dataset was imported from Google Drive in CSV format using Pandas.

```
pd.read_csv()
```

2. Initial Data Inspection

The dataset was initially explored to understand its structure, data types, and general characteristics using:

- `head()` to preview sample records
- `info()` to inspect data types and missing values
- `describe()` to generate descriptive statistics
- `shape` to identify the dataset dimensions

This step helped identify missing values, incorrect data types, and potential inconsistencies.

3. Fixing Data Types

The Timestamp column was originally stored as an object datatype. It was converted into datetime format using `pd.to_datetime()`.

After conversion:

- Separate Date and Time columns were extracted.
- The original Timestamp column was removed because it was no longer required for analysis.

```
pd.to_datetime()
```

4. Handling Missing Values

Missing values were identified using:

```
df.isnull().sum()
```

Different strategies were applied depending on the nature of the missing data:

a) Removing Unnecessary Columns

The Date and Time columns were dropped after inspection because they were not needed in the analysis.

b) Replacing Missing Text Responses

Missing values in open-ended survey questions such as:

- Stress effect opinions
- Sleep improvement methods

were replaced with:

'No Comment'

This ensured that unanswered questions were clearly distinguished from missing or corrupted data.

c) Handling Conditional Missing Values

Some missing values were caused by the questionnaire branching logic. Students who answered “No” to having insomnia or sleep disorders skipped the related follow-up questions.

To maintain logical consistency:

- Missing values in conditionally skipped questions were filled with appropriate default values.
- Some variables were filled using zeros to indicate absence of the condition.

5. Checking for Duplicates

Duplicate records were checked using:

```
df.duplicated().sum()
```

This step ensured that no repeated responses affected the statistical analysis.

6. Feature Encoding

Categorical sleep duration ranges were transformed into ordinal numerical values using ordinal encoding.

For example:

- Less than 4 hours → lower numerical value
- 7–8 hours → higher numerical value

This transformation enabled numerical analysis and visualization while preserving the natural order of sleep duration categories.

A mapping dictionary was used to perform the encoding process.

7. Logical Consistency Checking

A logical validation step was performed to ensure consistency between related survey responses.

For example:

- Students who indicated that they do not suffer from insomnia or sleep disorders were checked to ensure they did not contain contradictory responses in follow-up insomnia-related questions.

This process improved data integrity and minimized misleading correlations.

8. Correcting Encoding Errors

After encoding the sleep duration variable, some unmapped or missing values remained.

To resolve this issue:

- Missing encoded values were replaced using the mode (most frequent value).

```
.mode().iloc[0]
```

This ensured that the encoded feature contained no missing values.

9. Final Dataset Validation

After preprocessing:

- Data types were rechecked using `info()`.
- Missing values were verified again.
- The cleaned dataset was exported as:

`cleaned_sleep_dataset.csv`

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) was conducted to identify trends, patterns, and relationships between sleep habits, stress levels, and academic performance.

Both descriptive statistics and data visualization techniques were used.

1. Descriptive Statistics

Basic descriptive statistics were generated using:

`df.describe()`

This provided insights into:

- Distribution of encoded sleep duration
- Central tendency measures
- Data variability

2. Gender Distribution Analysis

A count plot was created to analyze the gender distribution of respondents.

Visualization used:

`sns.countplot()`

3. Sleep Quality Distribution

A pie chart was generated to visualize how students rated the effect of sleep quality on academic performance.

Visualization used:

`plot(kind='pie')`

4. Sleep Duration Distribution

A histogram with Kernel Density Estimation (KDE) was used to analyze the distribution of sleep duration among students.

Visualization used:

`sns.histplot()`

5. Sleep Disorder Distribution

Pie charts were used to examine the proportion of students suffering from insomnia or sleep disorders.

Main Research Question Analyses:

Several analyses were conducted to answer the study research questions.

I. Relationship Between Sleep Quality and Academic Performance

Is there a relationship between sleep quality and academic performance?

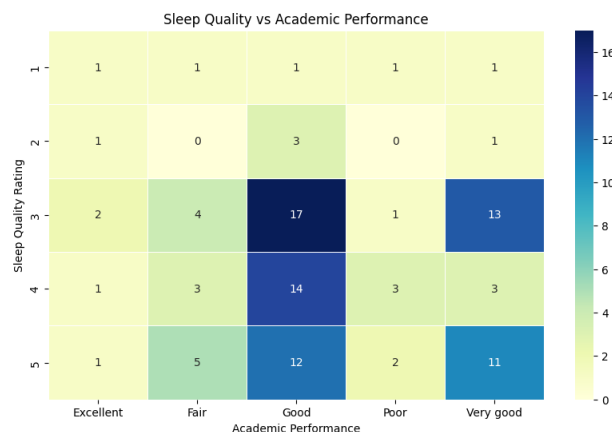
Variables

- Independent Variable (IV): Sleep Quality
- Dependent Variable (DV): Academic Performance

Data Visualization and Analysis

A cross-tabulation table and heatmap were created to explore the relationship between sleep quality and academic performance using:

- `pd.crosstab()`
- `sns.heatmap()`



In addition, a Spearman Correlation Test was conducted to measure the strength and significance of the relationship between the two variables.

Statistical Hypotheses

- H0: There is no significant relationship between sleep quality and academic performance.
- H1: There is a significant relationship between sleep quality and academic performance.
-

Statistical Findings

The Spearman correlation analysis revealed a very weak negative relationship between sleep quality and academic performance ($r = -0.082$). However, the relationship was not statistically significant ($p = 0.413 > 0.05$). Therefore, the null hypothesis was not rejected.

Insight

Most students reported moderate to high academic performance regardless of their sleep quality levels. Although students with better sleep quality appeared to achieve slightly more stable academic outcomes, the statistical analysis suggests that the relationship between sleep quality and academic performance is weak and not statistically significant in this dataset.

II. Factors Affecting Students' Stress Levels

Research Question

What factors affect students' stress levels?

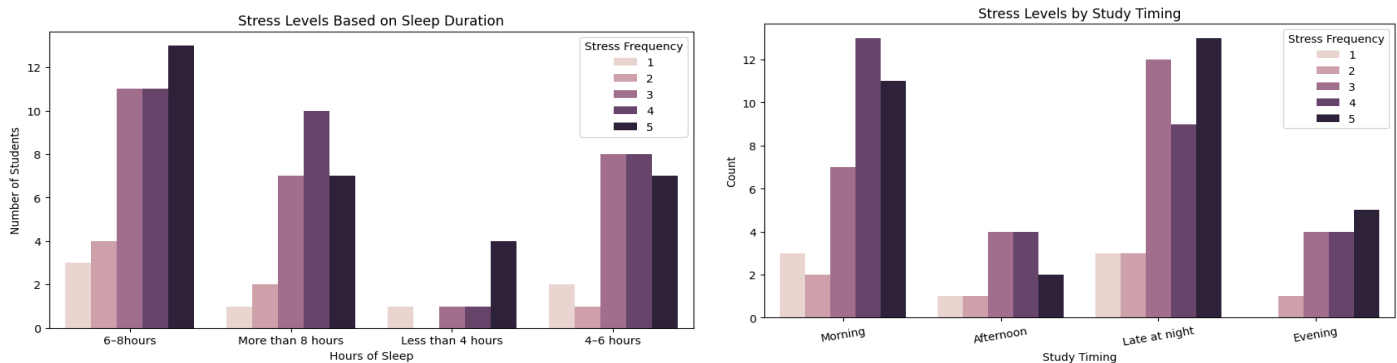
Variables

- Independent Variables (IVs): Sleep Quality, Sleep Duration, Study Habits
- Dependent Variable (DV): Stress Level

Data Visualization and Analysis

Countplots were created to visualize stress levels according to sleep duration and preferred study timing using:

- `sns.countplot()`



In addition, a Spearman Rank

Correlation was conducted to examine the relationship between sleep duration and stress levels, and a Kruskal-Wallis H-test was applied to compare stress levels across different study timing groups.

Statistical Hypotheses

- H0: There is no significant relationship between sleep duration and stress levels, and no significant difference in stress levels across study timing groups.
- H1: Sleep duration has a significant relationship with stress levels, or stress levels differ significantly across study timing groups.

Statistical Findings

Two statistical tests were conducted to examine the factors affecting students' stress levels. First, a Spearman Rank Correlation was used to assess the relationship between sleep duration and stress levels, revealing a negligible negative relationship ($r = -0.031$, $p = 0.761 > 0.05$). The result was not statistically significant, therefore the null hypothesis was not rejected. Second, a Kruskal-Wallis H-test was applied to compare stress levels across different study timing groups, showing no statistically significant difference ($H = 1.403$,

$p = 0.705 > 0.05$). Therefore, the null hypothesis was not rejected for this comparison either.

Insight

Students generally reported moderate to high stress levels regardless of differences in sleep duration or preferred study timing. Neither sleep duration nor study timing showed a significant relationship with stress levels in this sample, suggesting that students' stress is likely driven by other factors not captured in this study, such as academic workload, personal circumstances, or financial pressures.

III. Effect of Sleep Disorders on Academic Performance

Research Question

Do sleep disorders affect students' academic performance?

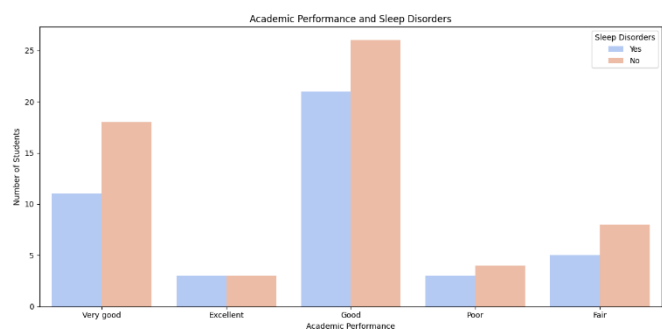
Variables

- Independent Variable (IV): Sleep Disorders
- Dependent Variable (DV): Academic Performance

Data Visualization and Analysis

A grouped countplot was created to compare academic performance levels between students with and without sleep disorders using:

- `sns.countplot()`



Additionally, an Independent Samples t-test was conducted to determine whether there was a statistically significant difference in academic performance between the two groups.

Statistical Hypotheses

- H_0 : There is no significant difference in academic performance between students with sleep disorders and students without sleep disorders.
- H_1 : There is a significant difference in academic performance between the two groups.

Statistical Findings

The Independent Samples t-test revealed no statistically significant difference in academic performance between students with sleep disorders and those without sleep disorders ($p = 0.984 > 0.05$). Therefore, the null hypothesis was not rejected.

Insight

Students with and without sleep disorders demonstrated similar academic performance distributions, with most students falling within the “Good” and “Very Good” categories. These findings suggest that sleep disorders did not significantly influence academic achievement among the surveyed students.

IV. Common Symptoms Associated with Sleeping Fewer Hours

Research Question

What are the most common symptoms associated with sleeping fewer hours?

Variables

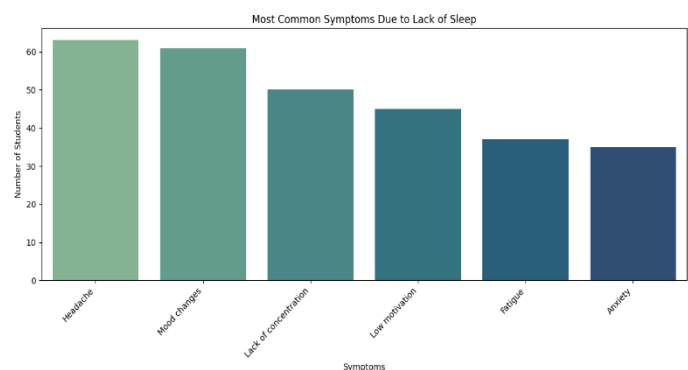
- Independent Variable (IV): Hours of Sleep
- Dependent Variable (DV): Sleep-Related Symptoms

Data Visualization and Analysis

Students were allowed to select multiple symptoms in the questionnaire. Therefore, symptoms were separated and analyzed individually to calculate their frequencies.

A bar chart was created using:

- `str.split()`
- `explode()`
- `sns.barplot()`



In addition, a Chi-Square Test of Independence was conducted to examine whether sleep duration was significantly associated with sleep-related symptoms.

Statistical Hypotheses

- H0: There is no significant association between sleep duration and sleep-related symptoms.
- H1: There is a significant association between sleep duration and sleep-related symptoms.

Statistical Findings

The Chi-Square test showed no statistically significant association between sleep duration and sleep-related symptoms ($p = 0.998 > 0.05$). Therefore, the null hypothesis was not rejected.

Insight

Headaches, mood changes, and lack of concentration were the most frequently reported symptoms among students experiencing insufficient sleep. However, the statistical analysis indicates that the occurrence of specific symptoms was not significantly associated with particular sleep duration categories.

V. Relationship Between Study Timing and Academic Performance

Research Question

Is there a relationship between study timing and academic performance?

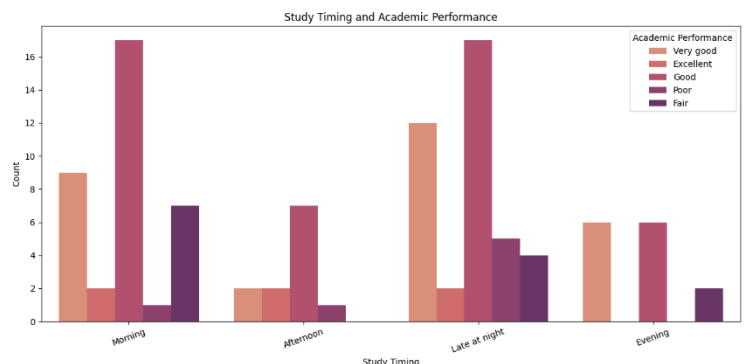
Variables

- Independent Variable (IV): Study Timing
- Dependent Variable (DV): Academic Performance

Data Visualization and Analysis

A grouped countplot was created to compare students' preferred study timing with their academic performance levels using:

- `sns.countplot()`



Additionally, a One-Way ANOVA test was conducted to determine whether academic performance significantly differed across study timing groups.

Statistical Hypotheses

- H0: There is no significant difference in academic performance based on study timing.
- H1: There is a significant difference in academic performance based on study timing.

Statistical Findings

The One-Way ANOVA test indicated no statistically significant difference in academic performance across different study timing groups ($F = 0.395$, $p = 0.757 > 0.05$). Therefore, the null hypothesis was not rejected.

Insight

Most students achieved “Good” and “Very Good” academic performance regardless of their preferred study timing. Although late-night studying showed slightly more variation in performance outcomes, study timing overall did not significantly affect academic achievement.

VI. Relationship Between Gender and Stress Level

Research Question

Is there a relationship between gender and stress level among students?

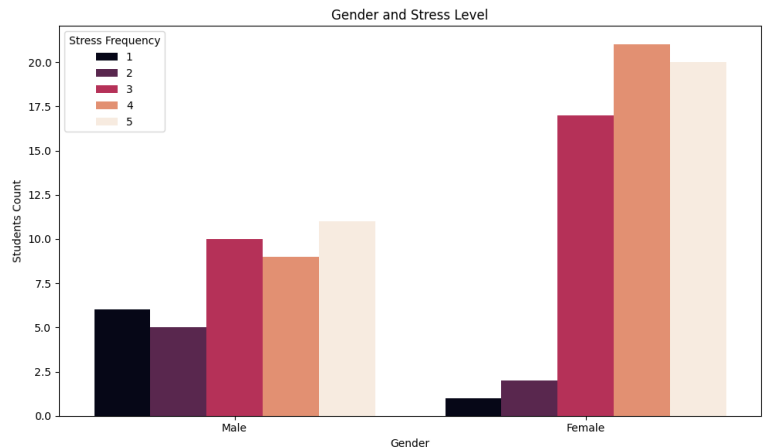
Variables

- Independent Variable (IV): Gender
- Dependent Variable (DV): Stress Level

Data Visualization and Analysis

A grouped countplot was created to compare stress levels across genders using:

- `sns.countplot()`



Additionally, an Independent Samples t-test was conducted to determine whether stress levels significantly differed between male and female students.

Statistical Hypotheses

- H0: There is no significant difference in stress levels between male and female students.
- H1: There is a significant difference in stress levels between male and female students.

Statistical Findings

The Independent Samples t-test revealed a statistically significant difference in stress levels between male and female students ($t = -2.566$, $p = 0.012 < 0.05$). Therefore, the null hypothesis was rejected, indicating that gender significantly affects stress levels among students.

Insight

The analysis suggests that stress levels vary significantly between genders. One gender group reported noticeably higher stress frequencies during the semester, indicating that gender-related factors may influence students' stress experiences.

Machine Learning Prediction of Academic Performance:

Objective

To predict students' academic performance using sleep, stress, study habits, and demographic-related variables.

Features Used

The following variables were used as input features:

- Sleep duration
- Sleep quality
- Stress level
- Insomnia status
- Study timing
- Gender

The target variable was:

- Academic Performance

Data Preparation

Categorical variables were encoded into numerical values using mapping techniques. The dataset was then divided into training and testing sets using:

- `train_test_split()`

Machine Learning Model

A Random Forest Classifier model was implemented using:

- `RandomForestClassifier()`

The model was trained to classify students' academic performance levels based on the selected features.

Model Evaluation

The model was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

Statistical Findings

The Random Forest Classifier achieved an accuracy of 23.81%, indicating limited predictive performance. The classification report showed that the model performed relatively better when predicting the "Good" academic performance category, while prediction performance for other categories remained weak.

The confusion matrix also demonstrated that the model struggled to correctly classify multiple academic performance levels.

Insight

The machine learning analysis suggests that academic performance cannot be accurately predicted using only the selected sleep and stress-related variables. Although some variables may contribute to academic outcomes, students' performance appears to be influenced by additional academic, psychological, and environmental factors not included in this study.

Conclusions & Recommendations

conclusion

The data shows that sleep quality, sleep duration, and study timing did not significantly affect students' academic performance or stress levels within this sample. However, gender played a clear role in stress levels, with female students reporting significantly higher stress frequencies than their male counterparts. These findings suggest that students' stress and academic outcomes are likely shaped by additional factors beyond those captured in this study. Universities should therefore focus on providing mental health support and promoting healthy sleep habits to help students maintain their well-being and academic success.

Recommendations

1. Although study timing did not show a statistically significant effect in this study, organizing study schedules during morning or daytime hours may help students maintain better sleep routines and reduce overall fatigue.
2. Promote Awareness of Sleep Duration and Quality:
Students must be advised against sleeping for less than 4 hours under any circumstances. It is important to emphasize that sleep quality is key to preventing common, debilitating symptoms like headaches, mood swings, and a lack of concentration and motivation.
3. Provide Targeted Mental Health Support:
Given the distinct disparity in stress levels, educational institutions should offer resources and programs for mental health support and stress management. Special attention and tailored support should be directed toward female students to help alleviate their disproportionately high stress burden.
4. Comprehensive Health Monitoring (Even with Sleep Disorders):
Since the presence of sleep disorders does not prevent students from achieving "Good" academic performance, the focus should shift toward supporting these students' physical and mental health to alleviate their fatigue. This ensures they can sustain their academic success without experiencing burnout.